Laboratory measurestream 001
Corso Duca degli Abruzzi, 24
10129 Torino, Italy

Tel. +39 02 3456 7890
e-mail calibration@measurestream.it
www.measurestream.it

CALIBRATION CERTIFICATE
Certificate of Calibration

2026/CAL/00419

Date of issue	2026-04-09
Customer	ThermoTech Industries S.p.A. | Via Industriale 88, 25030 Brescia (BS), Italy
Receiver	---
Request number	RQ-2026-0081
Request date	2026-03-28
Date of measurements	2026-04-07
Item	DIGITAL thermometer
Manufacturer	Measurestream electronics
Model	0001-3
Serial number	00010001-3

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Calibration procedure

PROCEDURE	Internal procedure validated by the quality service laboratory approved by Accredia based on the comparison method in a controlled oven with a reference, Pt100 platinum resistance thermometer read by Fluke 1502A metrology well.
TRACEABILITY	Procedure N. Traceability starts from first-line standards N. INRIM-PT100-REF-0042, FLUKE-1502A-REF-0017 with valid calibration certificates N. LAT-042-2025-0318, LAT-042-2025-0319 issued by INRIM - Istituto Nazionale di Ricerca Metrologica, Turin, Accredia LAT No. 042 - Measurestream S.r.l. with A.
Environmental calibration conditions	Temperature (23.0 +/- 1.5) °C Relative humidity (50 +/- 10) %
Calibration conditions	Calibration range: 0 °C to 125 °C Temperature steps: 0, 25, 50, 75, 100, 125 °C — 6 calibration points Reference: Fluke 1502A readout + Pt100 probe (U=0.065 °C, k=2) Sensor: DIGITAL thermometer (Measurestream electronics, model 0001-3), resolution 0.01 °C Settling period: 10 min per step

Additional notes

- Results refer to the instrument under calibration under the declared conditions.
- Uncertainties are determined according to ISO/IEC Guide 98-3 (GUM) and EA-4/02.
- Coverage factor k = 2.0, confidence level about 95 %.
- Reference instrument: Fluke / Hart Scientific FieldMetrologyWell_9142 (calibration certificate: 012-00000).
- Sensor under calibration: DIGITAL thermometer (Measurestream electronics, model 0001-3).
- $M_e = T_c - T_{ref}$ (signed temperature difference in °C).

This certificate is issued in compliance with the laboratory internal procedure PRO-CAL-MST-003. The results refer exclusively to the identified DIGITAL thermometer and to the declared calibration conditions. After having identified the F. taratura, the calibration function is: $T = A \cdot N + B$. The calibration was determined over 6 calibration points (0 °C to 125 °C) using a Fluke 1502A metrology well readout and Pt100 reference thermometer.

Statements

The reported results are valid only under the calibration conditions stated in this certificate.Partial or full reproduction is permitted only with written authorization from the laboratory.Expanded uncertainty is expressed with coverage factor k = 2 (confidence level about 95%).

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Calibration results

Object	DIGITAL thermometer
Model	0001-3
Serial number	00010001-3

Point	T_ref Reference temperature	T_c Measured value	M_e M_e (As found)	M_e residuals Measurement error (as-left)	U(E) Measurement Uncertainty
1	-20,00	-34,10	-14,10	1,5434	0,066
2	-10,00	-29,32	-19,32	-1,1497	0,071
3	-0,00	-21,98	-21,98	-1,3534	0,081
4	10,00	-11,56	-21,56	-0,0069	0,086
5	20,00	1,73	-18,27	1,0846	0,078
6	30,00	16,90	-13,10	0,8087	0,071
7	40,00	32,52	-7,48	-0,3418	0,071
8	50,00	47,37	-2,63	-0,9445	0,090
9	60,00	60,09	0,09	-0,6344	0,11
10	70,00	70,56	0,56	0,3519	0,076
11	80,00	78,78	-1,22	1,0480	0,086
12	90,00	85,10	-4,90	0,7239	0,074
13	100,00	89,81	-10,19	-1,1298	0,078

Notes

Temperatures are expressed in Celsius degrees according to the ITS-90 scale.
Reference temperature T_ref: mean of PT100 (Fluke 1502A) readings at each calibration step.
T_c: reading of the DIGITAL thermometer under calibration at each step.
M_e = T_c - T_ref (signed temperature difference in °C).
Measurement uncertainty U(E) is expanded (k=2), confidence level about 95 %.
Calibration was performed with measurement current 0.1 mA and a 2-terminal connection.

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Calibration function T/°C = f(D/LSB) — cubic

After having identified the calibration function, the cubic polynomial calibration function is:

The following table lists the coefficients of the cubic calibration equation:

$$T = A + B \cdot D + C \cdot D^2 + E \cdot D^3$$

Parameter	Value
regression uncertainty	2,3 °C
A [°C]	-28,60
B [°C/LSB]	0,004052
C [°C/LSB^2]	-9,561·10^-8
E [°C/LSB^3]	1,084·10^-12

Approval

Executed by	Dott. Marco Esposito
Authorized by	Ing. Giovanni Ferrari
Signature name	Marco Esposito